Homework 3

China SOE Factor

15.439 — Quantitative Investment Management

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A deep analysis of SOE factor construction, critique of the proposed methodology, and an alternative framework grounded in Fama–MacBeth regressions and Bayesian shrinkage estimation.

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1 Question 1: Why SOEs Represent a Unique Risk Factor

1.1 The Economic Intuition

The core thesis is that State Owned Enterprises operate under a **fundamentally different objective function** than Privately Owned Enterprises. Where a POE's management maximises shareholder value (or at least some proxy of it), an SOE's management simultaneously serves the interests of the state, the Communist Party, its employees, and—only residually—its minority shareholders. This dual mandate creates a systematic risk that is *orthogonal* to traditional risk factors.

Formally, consider the objective functions:

POE:

$$\max \ V(\text{equity}) = \mathbb{E}\left[\sum_t \frac{\text{DCF}_t}{(1+r)^t}\right]$$

SOE:

$$\max \ \lambda_1 V(\text{equity}) + \lambda_2 W(\text{political objectives}) + \lambda_3 S(\text{social stability})$$

where $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and, critically, $\lambda_1 < 1$ for SOEs.

The wedge between these two objective functions is not captured by industry membership, book-to-price, momentum, or other standard factors because it represents a **governance risk premium**—a systematic discount that investors demand for holding securities whose cash flows can be redirected at the discretion of a non-market actor (the state).

1.2 Why This Risk Is Distinct from Known Factors

1. **Not subsumed by industry exposure.** SOEs and POEs coexist within the same GICS sectors. In financials, state-owned banks (ICBC, CCB, BOC, ABC) sit alongside privately-controlled banks (CMB, Ping An Bank). The SOE risk operates *within* sectors, not across them.
2. **Not captured by book-to-price.** While SOEs tend to trade at lower P/B ratios (the CSI Central SOE Composite Index has historically traded at a significant P/B discount to the POE index), this discount reflects a fundamentally different phenomenon than the standard value premium. The value factor captures distress risk and mean-reversion in fundamentals. The SOE discount captures *agency costs and political risk* that may never mean-revert—indeed, under Xi Jinping's administration, state influence has intensified, not diminished.
3. **Not captured by size.** Many SOEs are among the largest companies in the A-share market (PetroChina, ICBC, Kweichow Moutai). The SOE effect operates across the size distribution.
4. **Not captured by momentum or sentiment.** SOE-related policy shocks (e.g., SASAC's 2024 "market value management" initiative, or periodic "valuation with Chinese characteristics" campaigns) create return dynamics that are idiosyncratic to the SOE/POE divide and unrelated to price trends or news tone.

1.3 Empirical Evidence on SOE vs. POE Differences

Key Data Points

- **Profitability gap:** State firms in China's industrial sector have a return on assets (ROA) roughly *half* that of private firms, and the gap is widening.
- **Scale:** China had ∼362,000 SOEs as of 2022. Among listed A-shares, 1,279 SOEs represent ∼29% of MSCI China's index weight and 30% of average daily turnover.
- **Asset base:** Central SOEs' total assets exceeded 90 trillion yuan (∼$12T) in 2024, with total profits of ∼2.6 trillion yuan.
- **Rising dominance:** SOEs' share of aggregate market capitalisation among China's 100 largest listed firms rose from ∼31% in 2021 to ∼54% in 2024.

1.3.1 Governance: The "Double Entry, Cross Offices" Mechanism

Under the CCP's governance model (*shuangxiang jinru*, *jiaocha renzhi*), the Party committee secretary and the board chairman of an SOE are typically the same person. Most Party committee members also serve on the board and in senior management. Decisions are effectively pre-determined in Party committees before board meetings occur—a governance structure with no analogue in POEs.

1.3.2 Policy Tail Risk

SOEs face a unique class of tail risks: directed acquisitions of distressed assets, mandated employment maintenance, subsidisation of downstream industries, and Belt-and-Road investments with negative expected returns. These represent *call options written by minority shareholders to the state*—a contingent liability absent for POEs.

1.4 A Mathematical Framing of the Risk Premium

If we denote the return on stock i at time t as:

$$R_{i,t} \;=\; \alpha_i + \beta_{i,\text{MKT}}\,\text{MKT}_t + \beta_{i,\text{SMB}}\,\text{SMB}_t + \beta_{i,\text{HML}}\,\text{HML}_t + \cdots + \beta_{i,\text{SOE}}\,\text{SOE}_t + \varepsilon_{i,t}$$

The hypothesis is that $\beta_{i,\text{SOE}}$ is priced—there exists a non-zero risk premium λ_{SOE} such that:

$$\mathbb{E}[R_i] \;=\; \sum_k \beta_{i,k}\,\lambda_k \;+\; \beta_{i,\text{SOE}}\,\lambda_{\text{SOE}}$$

If the SOE factor were truly spanned by existing factors, then regressing SOE_t on MKT, SMB, HML, etc. would yield an intercept (alpha) of zero and residuals that are pure noise. The empirical question is whether this intercept is statistically and economically significant.

2 Question 2: Weaknesses of the Proposed Factor Construction

The proposal is:

> *Long the top-10-by-market-cap SOEs per sector (equally weighted within sector, market-cap weighted across sectors, capped at 25%) minus short the CSI 300.*

This construction has several critical flaws, enumerated from most to least severe.

Flaw 1: The Net Exposure Problem — "What Does This Factor Actually Measure?"

The CSI 300 *itself contains a large number of SOEs*. SOEs represent approximately 29% of MSCI China and a comparable or higher fraction of the CSI 300 (which tilts toward large-caps where SOEs are over-represented).

Therefore, the net SOE exposure is:

$$\text{Net SOE Exposure} \approx \underbrace{1.0}_{\text{long leg}} - \underbrace{w_{\text{SOE, CSI300}}}_{\text{short leg}}$$

If $w_{\text{SOE, CSI300}} \approx 0.50$, you are obtaining only $\sim$50 cents of SOE exposure per dollar of GMV. The residual on the short side is a contaminated mix of POE shorts and residual SOE longs.

The return stream actually represents:

- A diluted SOE premium (what we want, but attenuated),
- A large-cap tilt within SOEs (top 10 by market cap),
- A sector-bet residual (despite the cap, sector weights differ from the CSI 300),
- An active-share component from equal-weighting within sectors vs. market-cap weighting in the index.

Flaw 2: The 25% Sector Cap Creates Arbitrary Distortions

Chinese SOEs are heavily concentrated in financials, energy, utilities, and industrials. If the natural market-cap weight of SOEs in financials exceeds 25%, the cap mechanically *under-weights* the sector where SOE exposure is densest. The factor under-represents the very sector where the SOE phenomenon is most pronounced.

Thought experiment: What if you *remove* the cap and market-cap weight everything? Then the factor degenerates toward the market itself (since large SOEs are market-cap weighted and the short side *is* the market). The return stream approaches zero. This reveals that the construction is fundamentally ill-conceived—you cannot build a clean SOE factor as "big SOEs minus the market" because *big SOEs are the market*.

Flaw 3: Equal Weighting Introduces Unintended Bets

Equal-weighting the top 10 SOEs within each sector overweights smaller SOEs relative to the largest ones. In financials, equally weighting the 10th-largest SOE bank with ICBC creates a massive implicit *small-within-large* tilt—an uncompensated bet that adds noise.

Flaw 4: "Top 10 by Market Cap" Selection Bias

Selecting only the 10 largest SOEs per sector introduces survivorship and selection bias. These are the best-run, most politically connected "national champion" SOEs—disproportionately *central* SOEs supervised by SASAC.

The risk characteristics of a central SOE like PetroChina differ markedly from those of a provincial-level SOE in the same sector. By selecting only the apex of the SOE hierarchy, the factor *misses the fat tail of SOE risk*—precisely where explanatory power should be greatest.

Flaw 5: Investability and Capacity Constraints

- **Free float:** Total SOE market cap is ∼\$6.6T but free float is only ∼\$2.8T. State-held shares are locked. Investing "several billions" would constitute a very large fraction of free float in many constituents.
- **ADT mismatch:** The 30% ADT figure applies to the *entire* SOE universe, not the specific top-10-per-sector basket, which will have substantially lower ADT.
- **Trading time mismatch:** The CXSE ETF (the only tradeable ex-SOE product) trades in the US on a T+1 lag relative to Asia market close. There is no clean, tradeable POE proxy available in Asia hours.

Flaw 6: The Short Side Is Wrong in Principle

A proper factor should be **long SOEs, short POEs** (*ceteris paribus*). By shorting the market, you are:

(i) Partially shorting SOEs (since they are in the index),
(ii) Shorting POEs with heterogeneous exposures to other risk factors,
(iii) Creating a net market exposure $\neq 0$ (the long basket's β_{MKT} is unlikely to be exactly 1.0).

The factor return is contaminated by market beta, sector tilts, and size effects that have nothing to do with SOE risk.

3 Question 3: An Alternative Construction of the SOE Factor

3.1 Design Principles

A well-constructed SOE factor should satisfy four criteria:

1. **Purity:** Isolate the SOE/POE dimension while neutralising all other known risk factors.
2. **Breadth:** Use the full cross-section of SOEs and POEs, not just the top 10.
3. **Investability:** Be realistically tradeable at scale.
4. **Transparency:** Have clear, unambiguous construction rules.

3.2 Approach A: Fama–MacBeth Regression (Preferred)

Cross-Sectional Neutralised SOE Factor

Step 1 — Define SOE status as a continuous variable.

Let:

$$\mathrm{GOV}_i \;=\; \text{fraction of shares held by government entities for stock } i \;\in [0,1]$$

sourced from WIND, CSMAR, or CCXE databases. WisdomTree uses a 20% threshold for its CXSE index; we use the *continuous* variable for factor construction and reserve the binary classification for interpretability.

Step 2 — Universe selection.

Include all A-share listed companies with:

- ADT $>$ \$1M over trailing 20 days,
- Market cap above the 5th percentile.

This yields a universe of ∼3,500–4,000 stocks.

Step 3 — Cross-sectional regression.

At each rebalance date t, for all stocks $j = 1, \ldots, n$:

$$R_{j,t} \;=\; \gamma_0 + \gamma_{\text{IND}}^{\top}\mathbf{IND}_j + \gamma_{\text{SIZE}} \ln(\text{MCap}_j) + \gamma_{\text{BP}} \left(\frac{B}{P}\right)_j + \gamma_{\text{MOM}}\,\text{MOM}_j + \gamma_{\text{SOE}}\,\text{GOV}_j + \varepsilon_{j,t}$$

where $\mathbf{IND}_j$ is a vector of industry dummies.

The coefficient $\hat{\gamma}_{\text{SOE},t}$ is the **pure SOE factor return** at time t, orthogonalised to industry, size, value, and momentum.

This is run T times for $t = 1, \ldots, T$. The reported statistics are:

$$\bar{\hat{\gamma}}_{\text{SOE}} = \frac{1}{T}\sum_{t=1}^{T} \hat{\gamma}_{\text{SOE},t} \qquad \text{and} \qquad t\text{-stat} = \frac{\bar{\hat{\gamma}}_{\text{SOE}}}{\text{se}(\hat{\gamma}_{\text{SOE},t})}$$

This is precisely the Fama–MacBeth (1973) methodology discussed in Lecture 2, Table 3 of Fama–French (1992).

Why Fama–MacBeth Is Superior

- Extracts the **marginal** effect of SOE ownership, controlling for confounds.
- Uses the **entire cross-section**, not a cherry-picked subset.
- Produces a factor return that is *by construction* orthogonal to industries, size, value, and momentum.
- Can be directly used in the multi-factor risk model to measure PM exposures.
- Avoids the "shorting the market" problem entirely.

3.3 Approach B: Sorted Portfolio (For Investability)

If a tradeable portfolio representation is needed—e.g., for hedging PM exposures—use a sorted-portfolio approach:

1. At each rebalance, **within each GICS sector**, sort stocks into terciles (or quintiles) by GOV_i.
2. Within each sector–GOV bucket, **value-weight** stocks by free-float market cap.
3. The factor portfolio is: **Long** the top-GOV tercile, **Short** the bottom-GOV tercile.
4. By construction, this is **sector-neutral** (identical sector weights on both sides).

3.3.1 Critical Refinements

- **Free-float adjustment:** Weight by free-float market cap, not total. This ensures investability and avoids allocating to locked-up shares.
- **Turnover buffer:** Require stocks to cross from one tercile to another by a buffer (e.g., must move from top 40% to bottom 40% to switch sides) to control turnover.
- **Liquidity screen:** Exclude stocks with ADT below a threshold.
- **Rebalance frequency:** Monthly or quarterly, aligned with the update cadence of government ownership data.

4 Question 4: Measuring Stock-Level Exposure to the SOE Factor

Once we have constructed the SOE factor return series $F_{\text{SOE},t}$, we need to estimate each stock's **beta** (sensitivity / loading) to this factor.

4.1 Approach 1: Time-Series Regression

For each stock i, over an estimation window $[\,t-T,\ t-1\,]$:

$$R_{i,\tau} - r_{f,\tau} \;=\; \alpha_i + \beta_{i,\text{MKT}}\,\text{MKT}_\tau + \beta_{i,\text{SMB}}\,\text{SMB}_\tau + \beta_{i,\text{HML}}\,\text{HML}_\tau + \beta_{i,\text{SOE}}\,F_{\text{SOE},\tau} + \varepsilon_{i,\tau}$$

for $\tau = t-T, \ldots, t-1$. The estimate $\hat{\beta}_{i,\text{SOE}}$ is the stock's sensitivity to the SOE factor.

Estimation window:

- Rolling window of 60–120 trading days ($\sim$3–6 months).
- Consider exponentially-weighted regression with half-life $\sim$60 days to weight recent data more heavily while retaining longer history.

Statistical considerations:

- Check Variance Inflation Factor (VIF) for multicollinearity between SOE and other factors.
- Correct standard errors for heteroskedasticity (White) and serial correlation (Newey–West, $\sim$5 lags).

4.2 Approach 2: Characteristic-Based Exposure

For *risk management* purposes—the CIO's stated use case—a characteristic-based measure is often superior.

Why Characteristics Can Dominate Regressions

Regression betas require a long history. For stocks that recently IPO'd, underwent ownership changes, or have limited trading history, regression betas are either unavailable or extremely noisy.

Recall Prof. Rothman's interview question #5: "How would you estimate the market β for a company that IPO'd 2 days ago?"

Proposed characteristic-based exposure:

$$\hat{\beta}_{i,\text{SOE},t} \;=\; f(\text{GOV}_{i,t},\ \text{CENTRAL}_{i,t},\ \text{SECTOR}_{i,t},\ \text{SASAC}_{i,t})$$

where:

- $\text{GOV}_{i,t} \in [0,1]$: government ownership fraction (continuous).
- $\text{CENTRAL}_{i,t} \in \{-1,\ 0,\ +1\}$: $+1$ if central SOE, 0 if local/provincial, -1 if POE.

- $\text{SECTOR}_{i,t}$: sector-specific adjustment (SOE risk varies—more intense in "strategic" sectors).
- $\text{SASAC}_{i,t} \in \{0, 1\}$: direct SASAC supervision indicator.

A parsimonious specification:

$$\hat{\beta}_{i,\text{SOE},t} \;=\; \text{GOV}_{i,t} \;\cdot\; \left(1 + \delta_{\text{SECTOR}(i)}\right)$$

where $\delta_{\text{SECTOR}(i)}$ is calibrated from the cross-section to capture the fact that a 30% government stake in a bank carries different SOE risk than a 30% stake in a consumer goods company.

4.3 Approach 3: Hybrid — Vasicek / Bayesian Shrinkage (Recommended)

Optimal Estimator via Bayesian Shrinkage

The optimal approach—mirroring what BARRA, Axioma, and Northfield do—combines both methods:

$$\hat{\beta}^{\text{hybrid}}_{i,\text{SOE}} \;=\; w_i \;\cdot\; \hat{\beta}^{\text{reg}}_{i,\text{SOE}} \;+\; (1 - w_i) \;\cdot\; \hat{\beta}^{\text{char}}_{i,\text{SOE}}$$

where the stock-specific shrinkage weight is:

$$w_i \;=\; \frac{\sigma^2_{\text{char}}}{\sigma^2_{\text{char}} + \sigma^2_{\text{reg},i}}$$

This is Vasicek (1973) shrinkage. Stocks with long, reliable return histories receive more weight on their regression beta. Stocks with short or noisy histories receive more weight on the characteristic-based prior.

4.4 Application to the Multi-PM Platform

As CIO, you need this factor exposure for two distinct purposes:

(a) **Ex-ante risk management:** Before a PM trades, you want to know how the trade changes the pod's SOE factor exposure. This requires *real-time* estimates $\Longrightarrow$ **use the characteristic-based approach**, updated whenever ownership data changes.

(b) **Ex-post attribution:** After the fact, decompose each pod's P&L into systematic factor returns and alpha. This requires well-estimated betas $\Longrightarrow$ **use the hybrid shrinkage approach** for monthly risk model updates.

4.5 Modelling Regime-Dependent Factor Volatility

The SOE factor's beta is *not constant*. During periods of state intervention (the 2015 "national team" market rescue, COVID-zero lockdowns, SASAC valuation campaigns), the factor's behaviour shifts dramatically. A well-specified risk model should allow for regime-dependent factor volatility:

$$\sigma^2_{\text{SOE},t} = \omega + \alpha F^2_{\text{SOE},t-1} + \beta \sigma^2_{\text{SOE},t-1} + \gamma \, \text{POLICY\_SHOCK}_t$$

where POLICY_SHOCK_t is a dummy for major SASAC announcements, CCP Congresses, or policy regime changes. This GARCH-with-exogenous-variable specification captures the fat tails and regime shifts that make SOE risk both difficult to model and essential to get right.

5 Summary Comparison

Dimension	Proposed (HW)	Recommended
Long side	Top 10 SOEs by mkt cap per sector	Full SOE universe, liquidity-filtered
Short side	CSI 300 (contains SOEs!)	POEs matched by sector (or FM regression)
Weighting	EW within sector, MCW across, 25% cap	Free-float VW within sector terciles
SOE classification	Binary	Continuous (GOV $\in [0, 1]$)
Factor neutrality	Sector cap only	Sector, size, value, momentum neutralised
Exposure estimation	Not specified	Hybrid Vasicek shrinkage (reg. + char.)
Investability	Questionable at scale	Liquidity-screened, free-float adjusted

Table 1: Comparison of the proposed and recommended SOE factor constructions.

This analysis draws on Fama–French (1992), the Fama–MacBeth (1973) regression framework, Vasicek (1973) Bayesian shrinkage, BARRA-style fundamental factor models, and the SOE/POE empirical literature. All factor construction recommendations are designed to be implementable with standard data sources (WIND, CSMAR) and tools available to a typical multi-manager quantitative platform.